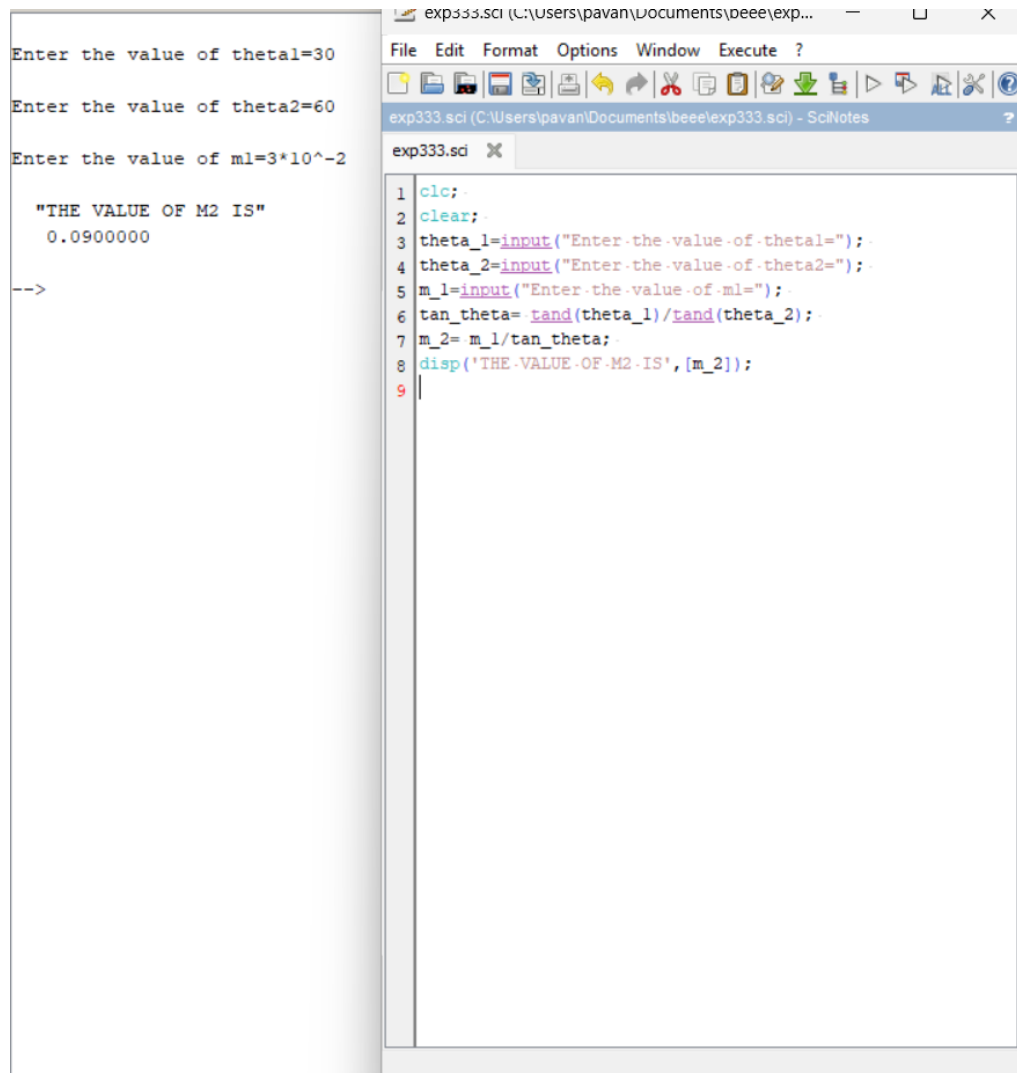


Exp no:10 Implement Boundary condition between two different dielectric medium in Magnetic Field and find the relative permeability of the different medium.



```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
File Edit Format Options Window Execute ?
exp333.sci X
1 clc;
2 clear;
3 theta_1=input("Enter the value of theta1=");
4 theta_2=input("Enter the value of theta2=");
5 m_1=input("Enter the value of m1=");
6 tan_theta= tand(theta_1)/tand(theta_2);
7 m_2= m_1/tan_theta;
8 disp('THE VALUE OF M2 IS',[m_2]);
9
```

Enter the value of theta1=30

Enter the value of theta2=60

Enter the value of m1=3*10^-2

"THE VALUE OF M2 IS"

0.0900000

-->

Calculation:

Exp 10

$\theta_1 = 30^\circ$
 $\theta_2 = 60^\circ$
 $m_1 = 8 \times 10^{-2}$

i) Calculate $\tan(\theta_1)$ and $\tan(\theta_2)$:-
 $\tan(30^\circ) = 0.577$, $\tan(60^\circ) = 1.732$

ii) Calculate ratio $\frac{\tan(\theta_1)}{\tan(\theta_2)}$:-
 $\frac{\tan(30^\circ)}{\tan(60^\circ)} = \frac{0.577}{1.732} = 0.333$

iii) For m_2 :-
 $\frac{\tan(\theta_1)}{\tan(\theta_2)} = \frac{m_1}{m_2} \Rightarrow m_2 = \frac{m_1}{\tan(\theta_1)/\tan(\theta_2)}$
 $m_2 = \frac{8 \times 10^{-2}}{0.333}$
 $m_2 = 9 \times 10^{-2} \approx 0.09$

Case 1:

```
% Exp 10
// Scilab code to simulate the magnetic field and flux density for two identical media
1
2
3 // Constants
4 mu_0=4*pi*10^(-7); // Permeability of free space (H/m)
5 H1=10; // Magnetic field intensity in medium 1 (A/m)
6
7 // Relative permeability for both media (both are 1)
8 mu_r1=1; // Relative permeability for medium 1
9 mu_r2=1; // Relative permeability for medium 2
10
11 // Calculate the magnetic field intensity in both media (no change because mu_r = 1)
12 H2=H1; // Since the media are identical, H1 = H2
13
14 // Magnetic flux density in both media (B = mu * H)
15 B1=mu_0*mu_r1*H1; // Magnetic flux density in medium 1
16 B2=mu_0*mu_r2*H2; // Magnetic flux density in medium 2
17
18 // Display results
19 disp("Magnetic Field Intensity in Medium 1: "+string(H1)+" A/m");
20 disp("Magnetic Field Intensity in Medium 2: "+string(H2)+" A/m");
21
22 disp("Magnetic Flux Density in Medium 1: "+string(B1)+" T");
23 disp("Magnetic Flux Density in Medium 2: "+string(B2)+" T");
24
25
```

Output:

```
Scilab 6.0.0 Console

"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 10 A/m"
"Magnetic Flux Density in Medium 1: 0.0000126 T"
"Magnetic Flux Density in Medium 2: 0.0000126 T"

-->
```

Case 2:

```
*exp 7.sc | exp6 case1.sc | exp6 case1.sc | exp6 case1.sc | exp7 c1.sc | exp c2.sc | exp8 c2.sc | exp8 c3.sc | exp9.sc | exp9 c3

1 // Scilab code to simulate the magnetic field intensity and flux density across the boundary
2
3 // Constants
4 mu_0=4*pi*10^(-7); // Permeability of free space (H/m)
5 H1=10; // Magnetic field intensity in medium 1 (A/m)
6
7 // Relative permeability for the media
8 mu_r1=4; // Relative permeability for medium 1
9 mu_r2=1; // Relative permeability for medium 2 (vacuum)
10
11 // Calculate the magnetic field intensity in medium 2 using the boundary condition
12 H2=mu_r1*H1/mu_r2; // H2 = (mu_r1 * H1) / mu_r2
13
14 // Calculate the magnetic flux density in both media (B = mu * H)
15 B1=mu_0*mu_r1*H1; // Magnetic flux density in medium 1
16 B2=mu_0*mu_r2*H2; // Magnetic flux density in medium 2
17
18 // Display results
19 disp("Magnetic Field Intensity in Medium 1: "+string(H1)+" A/m");
20 disp("Magnetic Field Intensity in Medium 2: "+string(H2)+" A/m");
21
22 disp("Magnetic Flux Density in Medium 1: "+string(B1)+" T");
23 disp("Magnetic Flux Density in Medium 2: "+string(B2)+" T");
24
25 |
```

Output:

```
"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 40 A/m"
"Magnetic Flux Density in Medium 1: 0.0000503 T"
"Magnetic Flux Density in Medium 2: 0.0000503 T"

--> |
```

Case 3:

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci
1 clc;
2 clear;
3
4 // Constants
5 mu_0 = 4 * pi * 1e-7; // Permeability of free space (H/m)
6 H1 = 10; // Magnetic field intensity in medium 1 (A/m)
7
8 // Relative permeability for the media
9 mu_r1 = 0.3; // Relative permeability for medium 1
10 mu_r2 = 2; // Relative permeability for medium 2
11
12 // Boundary condition (normal component of B is continuous)
13 H2 = (mu_r1 / mu_r2) * H1; // Magnetic field intensity in medium 2
14
15 // Magnetic flux density in both media (B = mu * H)
16 B1 = mu_0 * mu_r1 * H1; // Magnetic flux density in medium 1
17 B2 = mu_0 * mu_r2 * H2; // Magnetic flux density in medium 2
18
19 // Display results
20 disp("Magnetic Field Intensity in Medium 1: " + string(H1) + " A/m");
21 disp("Magnetic Field Intensity in Medium 2: " + string(H2) + " A/m");
22
23 disp("Magnetic Flux Density in Medium 1: " + string(B1) + " T");
24 disp("Magnetic Flux Density in Medium 2: " + string(B2) + " T");
25
```

Output:

```
"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 1.5 A/m"
"Magnetic Flux Density in Medium 1: 0.0000038 T"
"Magnetic Flux Density in Medium 2: 0.0000038 T"

-->
```